PASCAL
Private Secondary School Lemesos

Revision for Test 1 / Term 1

(Year 1-1A/B/C/D/E)

2026-2027

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Chapter 1: Analysing and displaying data 1.1 Average and range. 1.2 More averages and range. 1.3 Two-way tables and bar charts. 1.4 More graphs and tables 1. ~~1.5 More graphs and tables 2~~ 1 Check up 1 Strengthen 1 Extend 1 Unit test	**Unit 2 Rules of divisibility** 2.1 Rules of divisibility 2.2 Factors, multiples and primes 2.3 Positive and negative numbers 2.4 Squares and square roots 2.5 More powers and roots 2.6 Calculations 2 Check up 2 Strengthen

These revision notes **do not** intend to replace the material done in class. For a thorough revision you need to look back in the notes from class.

Exercise 1 – Answers

a) What is the average in a set of data called?

The mean is the average of a set of data. It is calculated by adding all the numbers and dividing by how many numbers there are.

b) What is the middle number when the data is arranged from least to greatest called?

The median is the middle number when the data is arranged in order.

c) What is the range of a set of data?

The range is the difference between the highest and the lowest numbers. You calculate it by subtracting the lowest number from the highest number.

d) What is the mode?

The mode is the number that appears most often in a set of data.

e) What is the only even prime number?

The only even prime number is **2**.

f) What is the highest common factor (HCF)?

The HCF is the largest number that divides exactly into two or more numbers.

g) What is the lowest common multiple (LCM)?

The LCM is the smallest positive number that is a multiple of two or more numbers.

h) What is a prime number?

A prime number is a whole number greater than 1 that has exactly two factors: 1 and itself.

Exercise 1

a) What is the average in a set of data called?

……

……

b) What is the middle number when the data is arranged from least to greatest called?

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c) What is the range of a set of data?

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d) What is the mode?

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e) What is the only even prime number?

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f) What is the highest common factor number?

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g) What is the lowest common multiple?

……

……

h) What is a prime number?

……

……

Exercise 2

Exercise 2 – Answers

a) 3, 6, 9, 12, 15, 18

b) 5, 10, 15, 20, 25

c) 2, 3, 5, 7, 11, 13, 17 prime

Write down the next two numbers in each sequence:

a) 3, 6, 9, 12, ... , ...

b) 5, 10, 15,..... , ...

c) 2, 3, 5, 7, 11, ... , ...

Exercise 3

common smaller

Find the HCF of each pair of numbers

a) 20 and 36

$20 = 2^2 \cdot 5^1$ (1) (2)

$36 = 2^2 \cdot 3^1$ (1)

b) 5 and 15

$5 = 5^1$

$15 = 3^1 \cdot 5^1$

all higher power

Exercise 4

Find the LCM of each pair of numbers

c) 56 and 98

$56 = 2^3 \cdot 7$

$98 = 2 \cdot 7$

$2 \cdot 7 = 8 \cdot 49 = 392$

d) 21 and 28

$21 = 3 \cdot 7$

$28 = 2^2 \cdot 7$

$= 2 \cdot 3 \cdot 7$

$= 4 \cdot 3 \cdot 7$

$= 84$

Exercise 5

At Dunkin Donuts ~~the~~ number of donut holes in a bag can vary. Help Donald find the mode.
12,10,10,10,13,12,11,13,10.

$= 10$

Exercise 6

Here are the weights, in kg, of 8 people.

63 65 65 72 74 86 90 97

a) Work out the median of the 8 weights.

b) Work out the range of the weights and the mode.

c) Work out the mean.

Exercise 7

The number of pages that Carolyn wrote in her journal each day from Monday to Friday is shown below:

6, 8, 12, 6, 10

a) Work out the median.

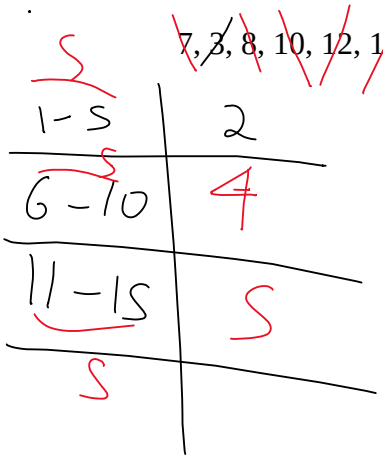
b) Work out the range.

d) Work out the mean.

Exercise 8

Write these values into a grouped frequency table

7, 3, 8, 10, 12, 12, 13, 14, 15, 5, 10.



Exercise 9

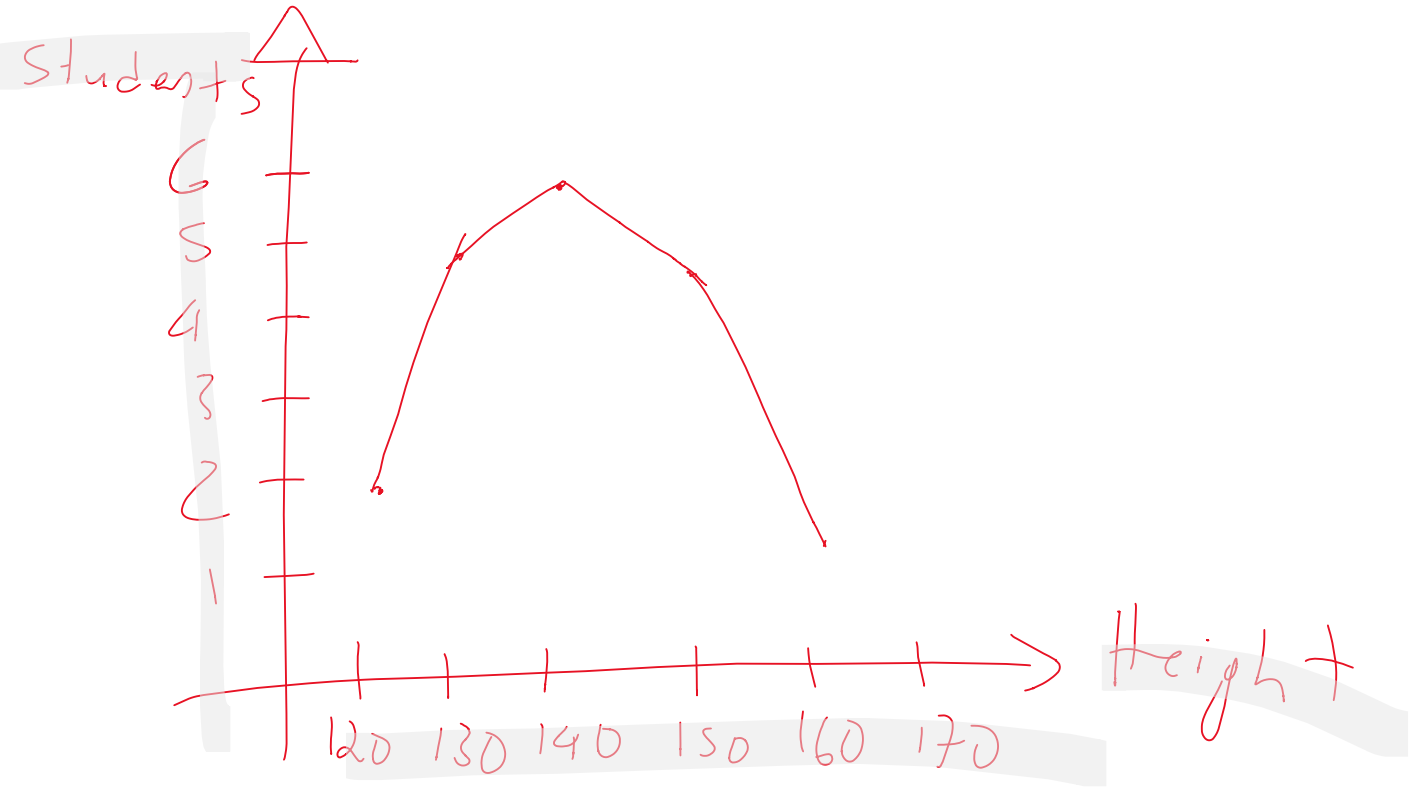
Andreas measured the heights, h cm, of some students in his class. The grouped frequency table shows his results

Height, h (cm)	Frequency
$120 \leq h < 130$	2
$130 \leq h < 140$	5
$140 \leq h < 150$	6
$150 \leq h < 160$	5
$160 \leq h < 170$	2

a) Make the frequency diagram

b) Which is the modal class? $140 \leq h < 150$

c) How many students are at least 150cm tall?



Exercise 10

A factory makes three sizes of bookcase. The sizes are small, medium and large. Each bookcase can be made from pine or oak or yew. The two-way table shows some information about the number of bookcases the factory makes in one week.

	Small	Medium	Large	Total
Pine	7	12	4	23
Oak	10	16	8	34
Yew	3	8	2	13
Total	20	36	14	70

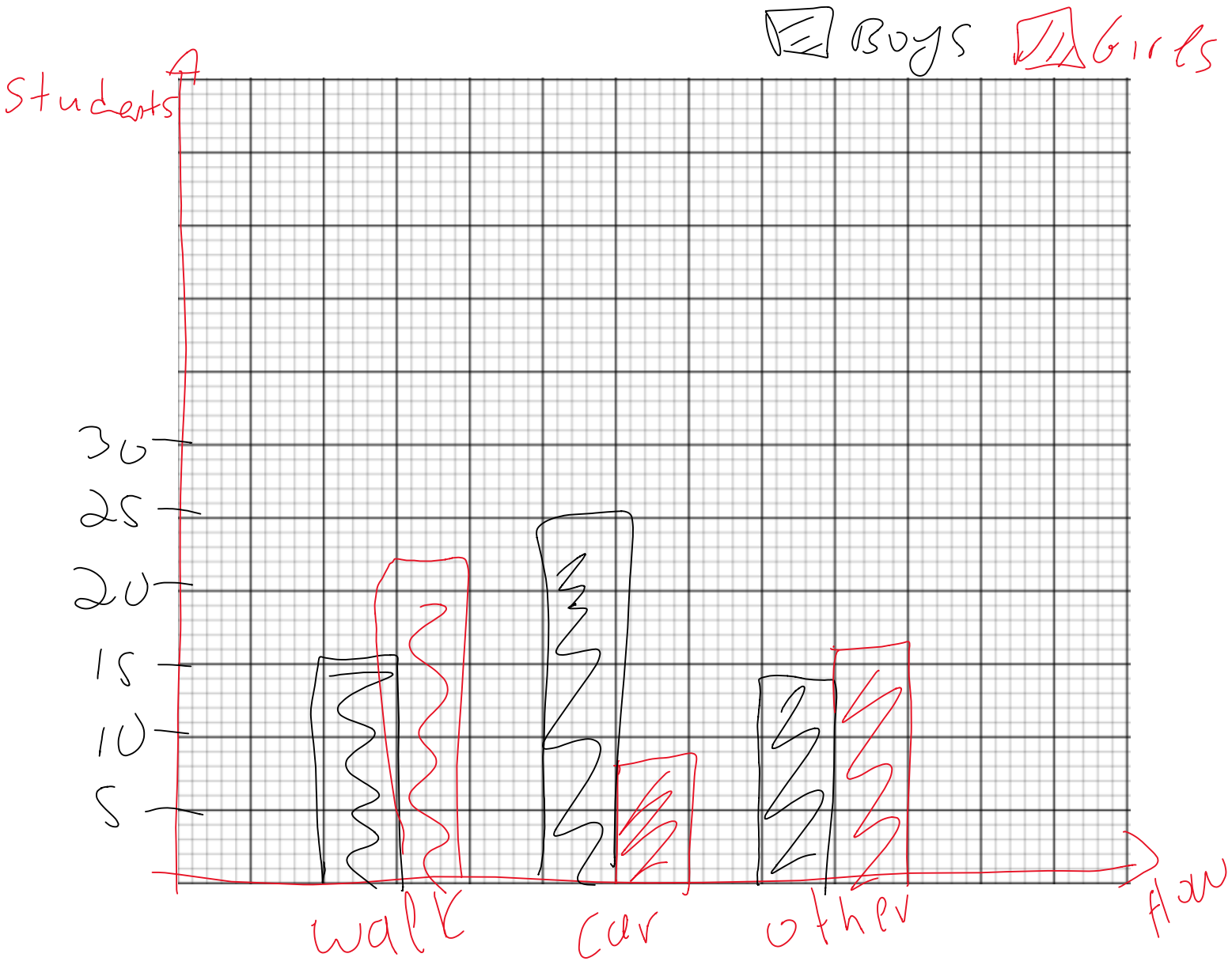
Complete the two-way table

Exercise 11

The two-way table gives some information about how 100 children travelled to school one day.

	Walk	Car	Other	Total
Boy	15	25	14	54
Girl	22	8	16	46
Total	37	33	30	100

a) Complete the two-way table.
b) Create the dual bar chart.



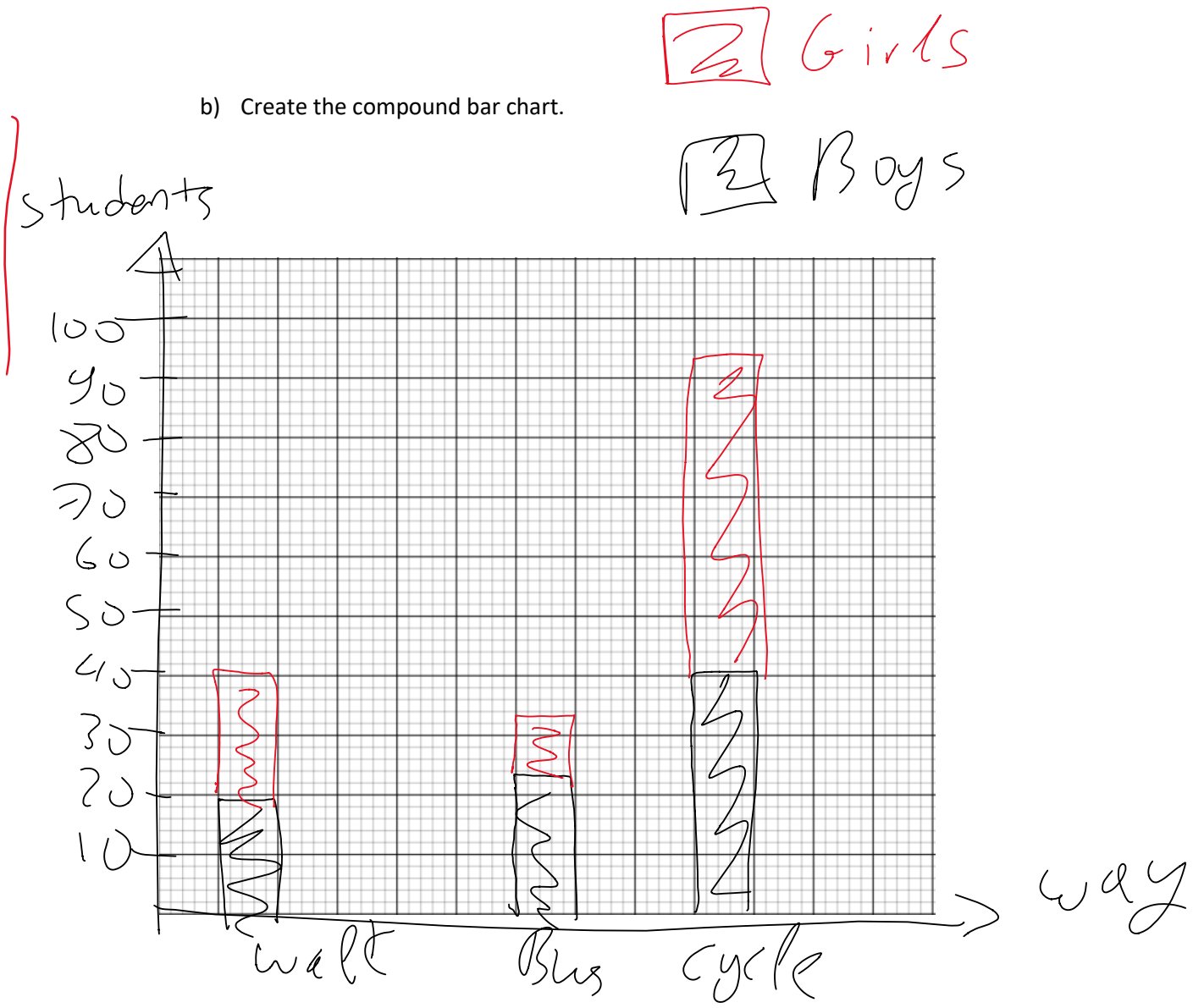
Exercise 12

There are 170 students in year 7 at a school. All of these students either walk to school, get the bus to school or cycle to school. 82 of the students are boys. 33 of the students get the bus to school. 19 of the 41 students that walk to school are boys. 56 girls cycle to school.

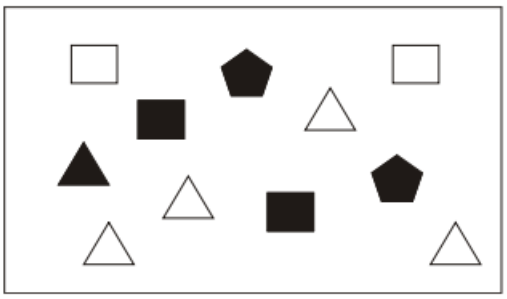
a) Complete the two way table.

	Walk	Bus	Cycle	Total
Boys	19	23	40	82
Girls	22	10	56	88
Total	41	33	96	170

b) Create the compound bar chart.



Exercise 13



The diagram shows some 3-sided, 4-sided and 5-sided shapes. The shapes are black or white.

Complete the two-way table.

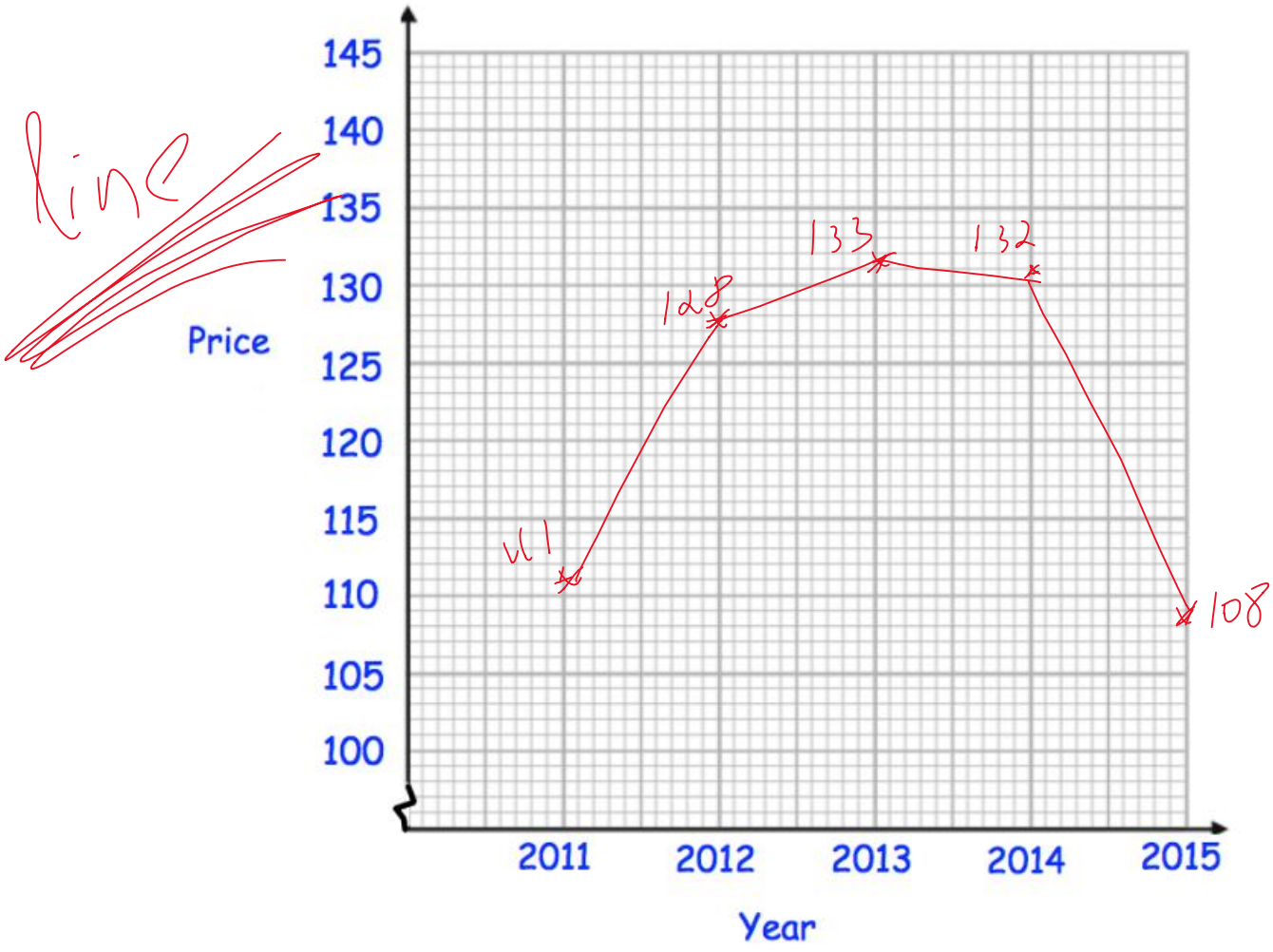
	Black	White	Total
3-sided shape	1	4	5
4-sided shape	2	2	4
5-sided shape	2	0	2
Total	5	6	11

Exercise 14

The table shows the average price of unleaded petrol in England over 5 years.

Year	Price in pence
2011	111
2012	128
2013	133
2014	132
2015	108

a) Draw a line graph for the data

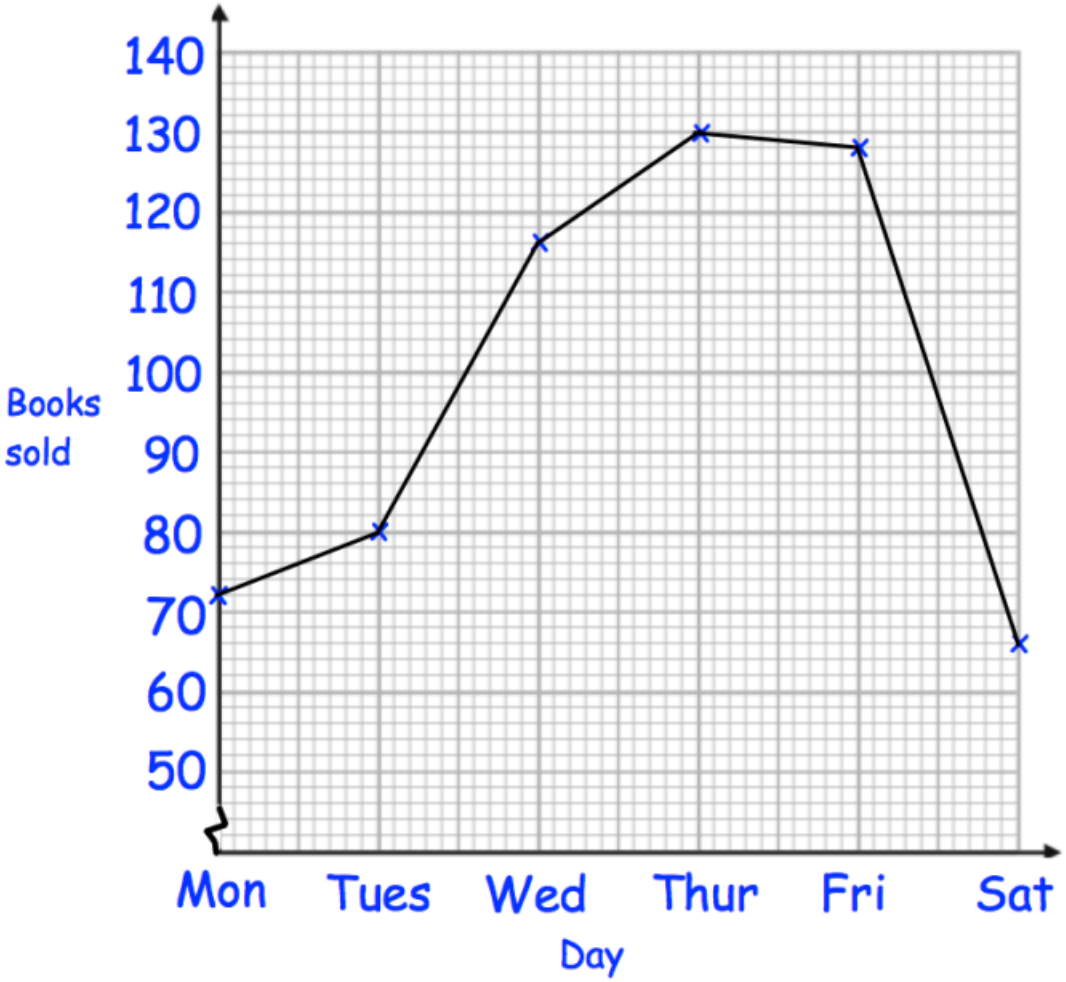


b) Between which two consecutive years did the price increase the most?

..... and .....

Exercise 15

Below is a line graph that shows how many books are sold in a charity shop over one week.



a) On which day did the charity shop sell the most books? ..... Thursday

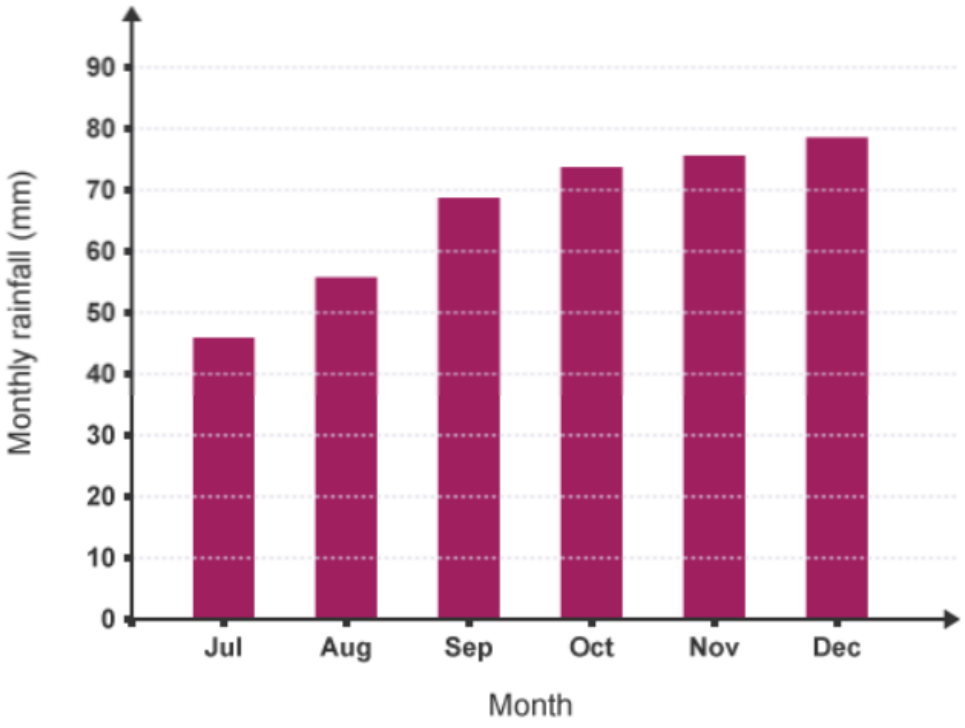
b) On which day did the charity shop sell the least books? ..... Saturday.

c) How many books were sold on Tuesday? 80

Exercise 16

What is misleading about this graph and statement?

Global warming warning! The ice caps are melting and rainfall is increasing. It is set to continue to rise from now on.



Exercise 17

a) Write down all the factors of 8

1 x 8 1, 8, 2, 4

2 x 4

b) Write down all the factors of 24

12, 1, 4, 2, 6, 8, 3, 24

c) Write down the common factors of 8 and 24

2, 8, 4, 1

Exercise 18

Write < (less than) or > (greater than) in the blank so that the expression is true.

(a) 7 ___ − 5

(b) −15 ___ 6

(c) −5 ___ − 20

(d) −3 ___ 3

(e) -7 ___ 0

Exercise 19

Find:

(a) −1+5 =

(b) 5+−6 =

(c) 15+−18−+9 =

(d) −14− −17 =

(e) (+3)×(+5) =

(f) (−8)×(+9) =

Exercise 20

Find the value of

(a) $9^2 =$ 81 9.9

(b) $4^3 =$ 4.4.4 = 64

(c) $30^2 =$ 30.30 = 900

(d) $0.6^2 =$ 0.6 x 0,6 = 0,36

(e) $\sqrt{81} =$ 9

(f) $\sqrt[3]{125} =$ 5

0.6
0,6
3 6
0 0 +
0,3 6

Exercise 21

Write these numbers as powers of 10

(a) 100 $= 10^2$

(b) 10000 $= 10^4$

(c) 100000 $= 10^5$

(d) 1000000 $= 10^6$

Exercise 22

Find the value of

(a) $2^3 \times 3 \times 2 =$ 8 x 6 = 48

(b) $2^2 \times 3^2 =$ 4 x 9 = 36

(c) $4^2 \times 2 =$ 16 x 2 = 32

Exercise 23

Write the following products in index form:

(a) $2\times2\times2 =$ 2^3

(b) $3\times3\times2\times3 =$ $3^3 \cdot 2$

(c) $7\times7\times7\times8\times4\times4 =$ $7^3 \times 8 \times 4^2$

Exercise 24

Work out an estimate for

(a) $\sqrt{46} =$ $\sqrt{36}=6$, $\sqrt{49}=7$; 6,6 6,7 6,8 6,5

(b) $\sqrt{89} =$ $\sqrt{100}=10$, $\sqrt{81}=9$; 9,4 9,3

(c) $\sqrt{50} =$ $\sqrt{64}=8$, $\sqrt{49}=7$; 7,1

Exercise 25

Work out:

(a) $\frac{1+\sqrt{81}}{\sqrt{100}} =$ $\frac{1+9}{10} = \frac{10}{10} = 1$

(b) $\frac{\sqrt[3]{64}+46}{5^2} =$ $\frac{4+46}{25} = \frac{50}{25} = 2$

(c) $\frac{8\times(11-\sqrt[3]{1000})}{\sqrt{4}} =$ $\frac{8\times(11-10)}{2} = \frac{8\times1}{2} = 4$